Considering Nonoverlapped Bands Construction: A General Dictionary Learning Framework for Hyperspectral and Multispectral Image Fusion

Yan Zhang, Lifu Zhang, *Senior Member, IEEE*, Ruoxi Song, Changping Huang, and Qingxi Tong

***Abstract*— Improving the spatial resolution of hyperspectral (HS) images is of great significance for the subsequent applications. As the multispectral (MS) image can provide abundant complementary land-cover spatial information, HS and MS image fusion (HMF) methods have become a mainstream to generate HS images with both high spatial and spectral resolution. HMF has witnessed rapid progress by leveraging dictionary learning (DicL) technique. However, existing approaches are highly sensitive to the image registration accuracy, and the reconstruction performance of the nonoverlapped spectral bands between HS and MS images is extremely limited. To alleviate the effect of image misregistration and enrich the spectral information of nonoverlapped bands, a general HMF DicL framework, which considers nonoverlapped spectral bands reconstruction and image misregistration, is proposed in this article. For registration error, the proposed method is rectified by the improved DicL, which can solve the problem of the spectral information matching gap existing in traditional HMF methods between HS image and MS image. Meanwhile, for nonoverlapped spectral bands reconstruction, a novel coefficient optimization strategy is adopted to improve the nonoverlapped bands reconstruction. Therefore, the registration error can be avoided to the greatest extent and the accuracy of nonoverlapped bands reconstruction can be effectively improved. Experiments both on simulated and real-world datasets demonstrate that the proposed method can effectively tackle the registration error problem and increase the HMF accuracy with different spectral ranges. Meanwhile, the proposed framework provides guidance significance for the DicL-based HMF methods with various constraints to improve the nonoverlapped bands reconstruction accuracy.**

***Index Terms*— Dictionary learning (DicL) framework, hyperspectral (HS) and multispectral (MS) image fusion (HMF), image misregistration, nonoverlapped spectral bands.**

I. Introduction

DUE to the hardware limitation of hyperspectral (HS) sensor, the captured HS image commonly has lower spatial resolution but higher spectral resolution than multispectral (MS) image and RGB image. To overcome this technical challenge, HS and MS image fusion (HMF) methods have become one of the most important means to generate HS images with both high spatial and spectral resolution [1], [2]. The purpose of HMF methods is to fuse the finer spatial information provided by MS or RGB image into HS image; in this way, the spatial resolution can be improved and the spectral resolution can be preserved simultaneously [3], [4]. It is worth noting that the RGB image can be seen as a special kind of MS image [5] and the RGB image is contained in the MS image concept mentioned in the following. Apparently, HMF is a more economical and more effective method than improving sensor hardware [6], [7]. The application potential of fused remote sensing image can also be largely broadened, such as sophisticated ground object recognition and refined resource survey [8], [9].

In terms of the fundamental methods used, HMF algorithms can be categorized into component substitution-based (CS-based) approaches [10], [11], probability statistics-based (PS-based) approaches [12], [13], matrix/tensor factorization-based (MF-based) approaches [14], [15], sparse coding-based (SC-based) approaches [16], [17], and deep learning-based (DL-based) approaches [18], [19].

The representative CS-based algorithms include wavelet transform-based methods [20] and pyramid-based and principal component transformation-based methods [21], [22]. The CS-based methods first emerge as panchromatic (PAN) and MS image fusion methods. Then, the CS-based methods are popularized and applied to MS and HS image fusion. With spectral bands of HS image increasing, spectral distortion may occur when implementing CS-based HMF methods. To ensure the best fusion performance of CS-based HMF methods, the spectral and spatial information of to-be-fused images must be highly correlated. Therefore, the CS-based methods need strict spectral and spatial matching.

Subsequently, Eismann and Hardie [23] presented a Bayesian-based fusion method named maximum a posteriori estimation and stochastic mixing model (MAP-SMM) fusion method, which is regarded as the foundational algorithm of PS-based HMF methods. Qi et al. [24] proposed a fast fusion algorithm based on Sylvester equation (FUSE) and it was also a representative PS-based HMF method. A typical difficulty in PS-based HMF methods is that the distribution of different images is diversified and it is difficult to describe uniformly.

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Due to the data complexity of HS image, CS- and PS-based HMF methods are deficient in handling HS image fusion.

The MF-based HMF methods represented by nonnegative matrix factorization (NMF) have been widely studied and applied due to their intuitive explanation of fusion process. The MF-based HMF methods usually build observation models of the fused image and the images to be fused through signal simulation and signal reconstruction theory. Then, some optimization solution methods can be implemented to compute the fused image. The representative MF-based methods are given, for example, Yokoya et al. [25] proposed a coupled NMF (CNMF) HMF method and Dian et al. [26] constructed a nonlocal sparse tensor factorization (NLSTF_SMBF) method.

The central idea of SC-based HMF methods is using finite feature vectors to reconstruct the original vector as much as possible. Among them, dictionary learning (DicL) and sparse representation (SR) are the most typical and most used SC methods [24]. The critical point in DicL-based methods is learning an overcomplete dictionary, which contains the main information of original image using dictionary to recover it. For example, an HMF method with optimized twin dictionaries (OTDs) [16] constructed two dictionaries to represent spatial and spectral features and then used them to achieve fusion. The SR-based methods, unlike DicL-based methods, emphasize using as few atoms as possible during the reconstruction process. For instance, Akhtar et al. [27] proposed an SR-based HMF method in which a spectra dictionary was extracted first, and then, the G-SOMP+ algorithm was used to solve the sparse coding problem and finished the fusion process.

With the development of computer vision, the DL-based HMF methods have attracted great attention and achieved great performance. The most commonly used network architectures include autoencoder (AE), convolutional neural network (CNN) [28], and generative adversarial network (GAN) [29], [30]. For example, Palsson et al. [31] used classic 3-D CNN architecture to achieve HMF purposes. These DL-based HMF methods using conventional network architectures ignore physical explanation during the HMF process and take fusion as simple image information mapping. Some researchers proposed some interpretable network architectures to relieve this issue. For instance, Xie et al. [32] first designed an exclusive MS and HS fusion network (MHF-net) for HMF, which embeds image degradation during framework designation. These interpretable DL-based HMF methods combine a quantitative description of imaging process and feature extraction of deep learning, making DL-based HMF methods further developed. On the other perspective, the DL-based HMF methods need adequate training data and training procedure to ensure that enough features are learned [33].

Existing HMF approaches have already achieved promising performance; however, they still face the following limitations. First, the abovementioned methods are implemented in a pixel-to-pixel fusion pattern, which requires that the original HS and MS images are fully geometrically registered [34]. Thus, to ensure the exact matching between the two images, the image registration strategy must be implemented before the image fusion task [35], [36]. During image registration, registration errors are accumulated and the fusion performance is highly sensitive to the registration accuracy. Second, the point spread function (PSF) and spectral response function (SRF) are adopted to simulate the image degradation process [31]. However, the SRF and PSF are difficult to acquire in many practical applications and need to be estimated [37], [38]. In this situation, their estimation accuracy will be significantly affected if there exists a large registration error. Third, the reconstruction performance of the nonoverlapped spectral bands between the HS image and the MS image is extremely limited. The HS image has wider and more continuous spectral bands than the MS image, and not all the bands contribute equally for different applications. Such the red edge band is sensitive to the vegetation monitoring task. Thus, completing the image fusion with different spectral ranges has important application significance. However, in the HMF process, the nonoverlapped spectral range does not have reference information in the MS image, which makes final image fusion more difficult. In conclusion, the accuracy of image registration and nonoverlapped spectral bands reconstruction are two key factors to determine the image fusion performance in real-world image fusion datasets [39].

To tackle the abovementioned limitations, in this article, a novel HS and MS image DicL fusion framework, which considers nonoverlapped spectral bands reconstruction and image misregistration (CNBM), is proposed. Compared with traditional HMF methods, the proposed framework discards pixel pairing fusion strategy and replaces it with end-member pairing. In this way, the misregistration influence of image can be effectively eliminated. From the spectral perspective, the high- and low-resolution HS images have the same spectral subspace. Thus, the proposed method extracts an HS dictionary to represent the spectral dimensional subspace of high spatial resolution HS (HRHS) image. Then, the HS dictionary is divided into overlapped and nonoverlapped spectral bands parts. The overlapped bands are responsible for linking information connections between MS and HS images. The transformation coefficient of nonoverlapped bands can be optimized by using nonoverlapped and entire HS dictionaries. Finally, the HRHS image can be reconstructed using two dictionaries (MS dictionary and nonoverlapped HS dictionary) and corresponding coefficient matrices. The major contributions of this article can be summarized as follows.

1) A general HMF DicL framework is proposed, which can effectively solve the image registration problem and improve the fusion performance of nonoverlapped spectral bands.
2) In the proposed framework, the end-member matching strategy is innovatively adapted to solve the image misregistration issue.
3) The proposed framework is comprehensively compared with some state-of-the-art (SOTA) HFM methods, and experimental results demonstrate the effectiveness of the proposed method in image misregistration and nonoverlapped spectral situations.
4) The proposed framework provides guidance significance for the DicL-based HMF methods with various

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constraints to improve the nonoverlapped bands' reconstruction accuracy.

The remaining content of this article is arranged as follows. Section II introduces the basic knowledge of related works in this article, including traditional expression model of HMF, DicL-based HMF method, and other basic algorithms involved. Followed by Section III, the principle and formula derivation of the proposed method are illustrated in detail. The validation and evaluation experiments are displayed and discussed in Section IV. The conclusion of this article is given in Section V.

II. Related Work

In this section, some related works are introduced, including general HMF expression model, basic knowledge of DicL-based HMF method, and related signal reconstruction algorithms.

A. Expression Model of HMF

In the DicL-based HMF methods, the fusion process can be regarded as an inverse process of HRHS image degradation. The low spatial resolution HS (LRHS) image is obtained by downsampling the spatial dimension of HRHS image using PSFs. Meanwhile, the high spatial resolution MS (HRMS) image is obtained by downsampling the spectral dimension of HRHS image using SRFs. On this basis, the degradation expression model of HRMS and LRHS can be depicted as

$$\boldsymbol{Z} = \boldsymbol{C}\boldsymbol{X} \tag{1}$$

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{R} \tag{2}$$

where, $\boldsymbol{Z} \in \mathbb{R}^{mn\times s}$, $\boldsymbol{Y} \in \mathbb{R}^{MN\times s}$, $\boldsymbol{X} \in \mathbb{R}^{MN\times s}$ represent the LRHS, HRMS and HRHS image, respectively, and m, n, S denote the spatial size and spectral bands number of LRHS; M, N, s denote the spatial size and spectral bands number of HRMS; $\boldsymbol{C} \in \mathbb{R}^{mn\times MN}$ and $\boldsymbol{R} \in \mathbb{R}^{s\times s}$ denote PSF and SRF, respectively.

B. HMF Based on DicL

In DicL frameworks, for a given sample, it can be represented by a linear combination of a coefficient matrix and an overcomplete dictionary, where the dictionary can encode the major semantic information of the images by learning a few training samples. Meanwhile, the coefficient matrix is used for looking up the exceptionally representative atoms of dictionary.

DicL-based HMF methods can be separated into the following three major steps: dictionary acquisition, coefficient matrix calculation, and recovery of the observed HRHS image. Pixel unmixing methods are the most commonly adopted strategy to obtain the atom dictionary.

On this basis, the reconstruction model of HRHS, LRHS, and HRHS image can be, respectively, depicted as

$$\boldsymbol{X}^T = \boldsymbol{D}_x\boldsymbol{H}_x + \boldsymbol{N}_x \tag{3}$$

$$\boldsymbol{Z}^T = \boldsymbol{D}_z\boldsymbol{H}_z + \boldsymbol{N}_z \tag{4}$$

$$\boldsymbol{Y}^T = \boldsymbol{D}_y\boldsymbol{H}_y + \boldsymbol{N}_y \tag{5}$$

Algorithm 1 OMP Algorithm

Input: a) x, observed vector to be reconstructed
b) $\boldsymbol{D}$, dictionary used in reconstruction
c) K, the number of atoms participated in reconstruction

Output: $\tilde{x}$, sparse approximation of x

Procedure:
(1) **Initialize:** residual $f^t = x$; index set $A = \phi$,
for $t = 1 : \mathrm{K}$
$\boldsymbol{d} = arg\min_d |< f^t, \boldsymbol{D}_t > |$
$\Lambda_t = \Lambda_{t-1} \cup \boldsymbol{d}$
$\boldsymbol{D}_t = \boldsymbol{D}_{t-1}/\boldsymbol{d}$
$\tilde{x}^t = \arg\min\left\|x - \boldsymbol{d}\tilde{x}^{t-1}\right\|$; $f^t = x - \tilde{x}$
end
return

where, $\boldsymbol{D}_x \in \mathbb{R}^{s\times k}$, $\boldsymbol{D}_z \in \mathbb{R}^{s\times k}$, $\boldsymbol{D}_y \in \mathbb{R}^{s\times k}$ are the basis spectral atom dictionaries of HRHS, LRHS, HRMS, respectively, k isnumber of spectral vectors of atom dictionary; $\boldsymbol{H}_x \in \mathbb{R}^{k\times MN}$, $\boldsymbol{H}_z \in \mathbb{R}^{k\times mn}$, $\boldsymbol{H}_y \in \mathbb{R}^{k\times MN}$ denote the coefficient matrix of HRHS, LRHS, HRMS, respectively; $\boldsymbol{N}_x, \boldsymbol{N}_z, \boldsymbol{N}_y$ represents the residual error during construction process of HRHS, LRHS, HRMS image, respectively.

To search for the optimal solution of coefficient matrices in (3)–(5), the object optimization function is constructed as

$$\boldsymbol{H}_x = \arg\min\|\boldsymbol{X} - \boldsymbol{D}_x\boldsymbol{H}_x\|_F^2. \tag{6}$$

To ensure the sparsity of coefficient matrix and further reduce the uncertainty of solution, sparse constraint is often added into (6) [40]. Therefore, the objection function can be written as

$$\boldsymbol{H}_x = \arg\min\|\boldsymbol{X} - \boldsymbol{D}_x\boldsymbol{H}_x\|_F^2 + \lambda\|\boldsymbol{H}_x\|_1 \tag{7}$$

where is the weighting coefficient of the sparse constraint, and it is related to the discrete degree of pixels in the to-be-fused image [41]. Thus, it can be determined by, $\lambda/\sigma = 0.14$, where σ is the standard deviation of the image to be fused.

To obtain the optimal solution of coefficient matrix, some signal compressed sensing reconstruction algorithm is used, such as the most commonly used orthogonal matching pursuit (OMP) algorithm. The OMP algorithm builds a linear sparse approximation of observed vector through constantly searching for atoms in dictionary. Each search is based on the residual of the previous search, which ensures the nonrepetition of the selected atoms and the rapidity of iterative solution. The detailed procedure of OMP algorithm is shown in Algorithm 1.

Each pixel in HRHS image is regarded as an observed vector, which can be recovered by the corresponding dictionary and coefficient matrix. The detailed solution method of coefficient matrix can be found in Section III.

C. Iterated Soft-Thresholding Algorithm (ISTA)

Equation (7) is a typical l_1-norm constraint optimization problem which can be abstractly summarized as

$$J(x) = \|y - x\|_F^2 + \lambda\|x\|_1. \tag{8}$$

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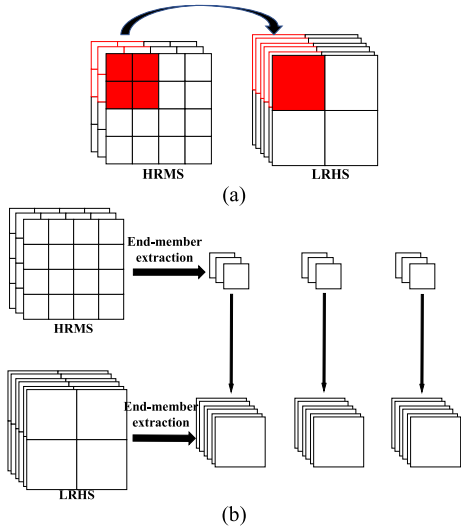

Fig. 1. Diagram of (a) image registration and (b) end-member matching.

Many algorithms have been developed for the purpose of minimizing (8). The ISTA is the earlier and most commonly used algorithm for optimizing such problem [42]. The optimized solution of (8) using ISTA can be achieved by

$$\text{soft}\left(y, \frac{\lambda}{2}\right) = \text{sign}(y)\max\left(0, |y| - \frac{\lambda}{2}\right) \tag{9}$$

where sign$(\cdot)$ denotes the sign function, which is used to determine whether the specified value is positive or negative. The detailed derivation process of (9) can be found in the literature [43], [44].

III. Proposed Method

In this section, we will discuss the proposed HMF DicL framework, which considers the nonoverlapped spectral bands reconstruction and image misregistration.

A. Observation Model of the Proposed Framework

The pixel level HMF methods require that the MS and HS image to be processed must be well coregistered. Fig. 1(a) shows the image registration process between the HS and MS images, and the pixels with the same geographical position should be one-to-one absolute correspondence to facilitate the subsequent fusion process. As the spatial registration process is extremely challenging, especially for submeter-level images. To alleviate the influence of inevitable registration error, the proposed framework adopts end-member matching instead of pixel-to-pixel matching strategy [as shown in Fig. 1(b)].

Based on the end-member matching strategy, the HS dictionary is further divided into overlapped HS dictionary, the spectral range of which is consistent with the HRMS image, and nonoverlapped HS dictionary, which spectral range is beyond of HRMS image. To determine the nonoverlapped spectral bands between HS and MS images, two fundamental technical parameters of these images should be investigated in advance, that is, the entire spectral range and the central wavelength of each band. The spectral range of two images can determine the nonoverlapped spectral range. The central wavelength of each band can determine which bands of LRHS locate in the nonoverlapped spectral range. These two parameters can be easily found in the image header file.

In this way, the expressing model of HRHS image can be reformulated as

$$\boldsymbol{X} = \left(\boldsymbol{D}_y \boldsymbol{H}_y\right)^T \boldsymbol{A} + \left(\boldsymbol{D}_{z'} \boldsymbol{H}_{z'}\right)^T \boldsymbol{B} \tag{10}$$

where $\boldsymbol{A} \in \mathbb{R}^{s \times S}$ and $\boldsymbol{B} \in \mathbb{R}^{s \times S}$($s'$ is the number of nonoverlapped bands) denote the connection matrix of overlapped and nonoverlapped spectral bands in HRHS image, respectively; $\boldsymbol{D}_{z'} \in \mathbb{R}^{s' \times k}$ is hyperspectral dictionary of nonoverlapped spectral bands and $\boldsymbol{H}_{z'} \in \mathbb{R}^{k \times MN}$ is coefficient matrix of nonoverlapped spectral bands construction.

Based on the improved expression model of HRHS image, the principal steps of the proposed method can be summarized in Fig. 2. There are three main stages of the proposed method: dictionaries construction, connection and coefficient matrices estimation, and HRHS reconstruction.

B. Proposed Methodology

Based on the idea of the proposed framework, the reconstruction model of MS image can be reformulated as

$$\boldsymbol{Y} = \boldsymbol{X}\boldsymbol{R} = \left[\left(\boldsymbol{D}_y \boldsymbol{H}_y\right)^T \boldsymbol{A} + \left(\boldsymbol{D}_{z'} \boldsymbol{H}_{z'}\right)^T \boldsymbol{B}\right]\boldsymbol{R}. \tag{11}$$

In (11), $\boldsymbol{D}_y$ and $\boldsymbol{D}_{z'}$ can be directly estimated by HRMS and LRHS image; $\boldsymbol{H}_y$ can be calculated through reconstructing HRMS image using OMP algorithm; connection matrix $\boldsymbol{A}$ and $\boldsymbol{B}$ can be estimated by three dictionaries, the specific estimation formula of matrix $\boldsymbol{A}$ and $\boldsymbol{B}$ can be found in (13)–(15); and finally, only $\boldsymbol{H}_{z'}$ is left unknown. Thus, the optimization problem of HMF can be designed as:

$$\min \left\| \left[\left(\boldsymbol{D}_y \boldsymbol{H}_y\right)^T \boldsymbol{A} + \left(\boldsymbol{D}_{z'} \boldsymbol{H}_{z'}\right)^T \boldsymbol{B}\right]\boldsymbol{R} - \boldsymbol{Y} \right\|_F^2 + \lambda f\left(\boldsymbol{H}_{z'}\right) \tag{12}$$

where $f(\boldsymbol{H}_{z'})$ represents the regularization constraint of $\boldsymbol{H}_{z'}$, which usually uses the l_1-norm as the sparse constraint.

Since matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are the connection matrices of overlapped and nonoverlapped bands between before and after fusion, they can be estimated by the corresponding dictionary matrices. The connection matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ can be computed by

$$\boldsymbol{A} = \left(\boldsymbol{D}_y^T \boldsymbol{D}_y\right)^{-1} \boldsymbol{D}_y^T \boldsymbol{D}_{z''} \tag{13}$$

$$\boldsymbol{D}_{z'} = \boldsymbol{U}\boldsymbol{S}\boldsymbol{V} \tag{14}$$

$$\boldsymbol{B} = \boldsymbol{V}(:, 1:l) \tag{15}$$

where $\boldsymbol{D}_{z'}$ is the dictionary matrix of overlapped bands; $\boldsymbol{U}$, $\boldsymbol{S}$, and $\boldsymbol{V}$ are the results of singular value decomposition of matrix $\boldsymbol{D}_{z'}$; and l is the number of nonoverlapped bands.

Then, the optimization (12) can be solved by ISTA. First, the auxiliary function is built to assist in searching the optimal

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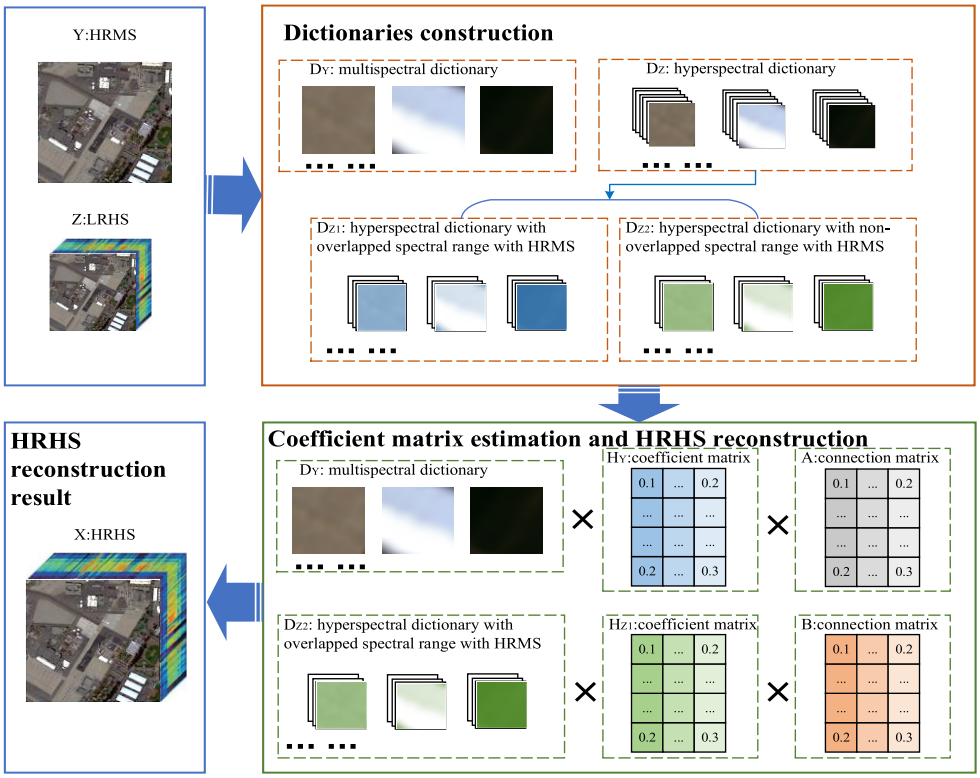

Fig. 2. Flowchart of the proposed framework.

solution

$$G\left(H_{z'}, H_{z'}^{(k)}\right) = g\left(H_{z'}^{(k)}\right) + < H_{z'} - H_{z'}^{(k)}, \nabla g\left(H_{z'}^{(k)}\right) > + \frac{1}{2t_k}\left\|H_{z'} - H_{z'}^{(k)}\right\|_F^2 + \lambda\|H_{z'}\|_1 \quad (16)$$

$$g\left(H_{z'}^{(k)}\right) = \left\|\left[\left(D_y H_y\right)^T A + \left(D_{z'} H_{z'}\right)^T B\right] R - Y\right\|_F^2 \quad (17)$$

where $g(H_{z'}^{(k)}) = D_{z'}^T\|[(D_y H_y)^T A + (D_{z'} H_{z'})^T B]R - Y]R^T B^T$ and t_k represents the step size.

As the introduced terms in (16) are nonnegative, the auxiliary function has the following two properties.

1) $G(H_{z'}, H_{z'}) = g(H_{z'})$.
2) $G(H_{z'}, H_{z'}) > g(H_{z'})$.

Based on these properties, the auxiliary function can be regarded as upper bound of primitive objection function and it is easier to optimize than the primitive function.

Omit the constant term in (16) and simplify it to

$$\frac{1}{2t_k}\left\|H_{z'} - \left(H_{z'}^{(k)} - t_k \nabla g\left(H_{z'}^{(k)}\right)\right)\right\|_F^2 + \lambda\left\|H_{z'}^{(k)}\right\|_1. \quad (18)$$

As (18) has the same form of (8), in which $H_{z'}$ replaces the position of x, thus, (18) can be solved by the following soft threshold method:

$$H_{z'}^{(k+1)} = \text{soft}\left[H_{z'}^{(k)} - t_k \nabla g\left(H_{z'}^{(k)}\right), \lambda t_k\right]. \quad (19)$$

Decomposing (19) in detail, the iterative solution of HFM can be derived as

$$X^{(k)} = \left(D_y H_y\right)^T A + \left(D_{z'} H_{z'}\right)^T B \quad (20)$$

$$E^{(k)} = Y - X^{(k)} B \quad (21)$$

$$G^{(k)} = D^{(k)} E^{(k)} R^T B^T \quad (22)$$

$$H_{z'}^{(k+1)} = \text{soft}\left[H_{z'}^{(k)} - t_k G^{(k)}, \lambda t_k\right]. \quad (23)$$

Once acquired, $H_{z'}$ the HRHS image X can be reconstructed.

IV. Experiments and Discussion

In this section, the performance of the proposed method, HMF method considering nonoverlapped spectral bands and image misregistration (CNBM), is evaluated on both real and simulated datasets. Three types of experiments are conducted in this article: validation experiments on simulated dataset, validation experiments on ZY1E HS and MS images, and RGB and HS image fusion on simulated unmanned aerial vehicle (UAV) image. Some representative algorithms are used to carry out comparison experiments, including CNMF [25], NLSTF_SMBF [26], adaptive Gram–Schmidt (GSA) spectral sharpening [45], HS super-resolution (Hysure) [46], an FUSE [24], HMF fusion method with OTDs [16], G-SOMP+ [27], and CNN-based fusion method (CNN-Fus) [28].

A. Validation Experiments on Simulated Dataset

The scene in the first dataset is Sandiego airport with a spatial size of 400 × 400 pixels, which was captured by the

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TABLE I
FUSION EVALUATION RESULTS OF NINE ALGORITHMS ON THE SANDIEGO AIRPORT DATASET

	SAM(↓,0)	RMSE(↓,0)	ERGAS(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)	Time/s
CNMF	4.176	407.464	4.154	39.651	0.842	31
NLSTF_SMBF	6.095	734.256	7.425	34.556	0.803	71
GSA	6.663	420.077	4.241	39.491	0.783	2
Hysure	7.307	616.321	6.204	36.219	0.675	180
FUSE	6.480	586.760	5.840	36.694	0.692	2
OTD	15.163	701.861	28.708	42.126	0.721	220
G-SOMP+	8.866	726.245	31.234	37.960	0.675	289
CNN-Fus	9.072	680.310	6.824	35.590	0.609	150
CNBM	**3.868**	**233.009**	**2.469**	**49.780**	**0.951**	29

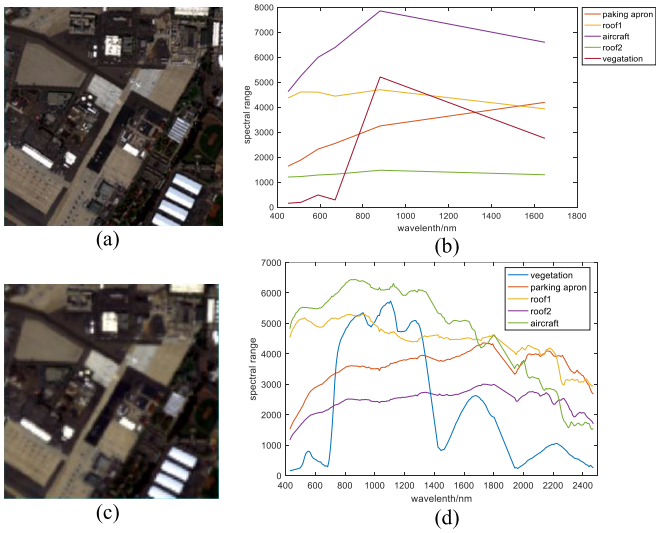

Fig. 3. (a) HRMS and (c) LRHS image of Sandiego airport dataset. (b) MS and (d) HS curves of major land-cover materials in the Sandiego airport dataset.

Airborne Visible InfraRed Imaging Spectrometer (AVIRIS). As shown in Fig. 3(a), the major land-cover object includes various roofs, parking apron, and aircraft. The MS and HS curves of the major materials in the Sandiego airport dataset are, respectively, shown in Fig. 3(b) and (d). The spectral range of the original HRHS image is in the range of 429–2470 nm, which is distributed with 189 spectral bands. The LRHS image is constructed by downsampling the original HRHS image into a spatial size of 100 × 100 pixels. Using SRF of first six bands of Operational Land Imager (OLI), the HRMS image [as shown in Fig. 3(a)] is simulated. Thus, the spectral bands of HRMS range from 443 to 1650 nm, which covers visible bands and part of near red bands. Totally speaking, the spectral range between HRMS and LRHS images in the first dataset is not exactly coincident.

To quantitatively evaluate the fusion performance of the fusion results, spectral angle mapping (SAM), root-mean-square error (RMSE), relative dimensionless global error in synthesis (ERGAS), peak signal-to-noise ratio (PSNR), and structural similarity (SSIM) are selected as the evaluation index. In addition, computation time is also used to evaluate the computational cost of all HMF methods. The quantitative evaluation results of Sandiego airport dataset are shown in Table I. The optimal tendency and value of assessment indexes are marked bold after indexes name. All the experiments were conducted with an Intel Core i9-10900K CPU with 3.70 Hz and 64-GB RAM PC.

As shown in Table I, all the indexes of the proposed method achieve the best performance among the comparison HMF methods. The PSNR and SSIM indexes demonstrate that the HRHS image generated by CNBM inherits more spatial information and image details. The other three indexes show that the fused image by CNBM preserves the spectral information well. From the fusion results of full spectral range of Sandiego airport dataset, the proposed method outperforms other algorithms. The time cost of CNBM algorithm spending on fusion is also within an acceptable range compared with other HMF methods. To further evaluate the fusion results of overlapped and nonoverlapped spectral bands, respectively, the fusion performance of the separated overlapped and nonoverlapped spectral bands of HRHS image is also evaluated. The corresponding evaluation results are shown in Table II.

As shown in Table II, another conclusion that can be observed is that in the overlapped spectral range, the proposed method can largely improve the fusion performance, and in the nonoverlapped spectral range, it has also relatively improved. Together overlapped and nonoverlapped spectral bands, the proposed method reaches a superior fusion performance than the other algorithms. However, the fusion effect of nonoverlapped spectral bands is universally worse than that of overlapped spectral bands in the Sandiego airport scene. There are two main reasons for this phenomenon. First, there is no reference information for nonoverlapped bands so that the SRF involved in the fusion process cannot be estimated accurately. Second, the mapping relationship between HRMS and LRHS image is relied on the overlapped spectral bands in that connection relationship between the nonoverlapped spectral bands in LRHS and HRHS is not distinct.

Followed by quantitative evaluation of fusion performance, additional experiments are designed to verify the resisting registration error function of nine algorithms. On account of accurate geometric coregistering is a vital factor on

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TABLE II
FUSION EVALUATION RESULTS OF OVERLAPPED AND NONOVERLAPPED SPECTRAL BANDS

	Overlapped spectral range			Non-overlapped spectral range		
	SAM(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)	SAM(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)
CNMF	3.189	41.729	0.913	4.861	38.958	0.838
NLSTF_SMBF	4.979	42.025	0.882	**4.378**	37.156	0.738
GSA	5.934	40.062	0.810	6.204	37.628	0.740
Hysure	5.954	36.747	0.725	7.646	34.266	0.580
FUSE	5.274	37.253	0.738	6.130	34.852	0.617
OTD	9.788	41.863	0.766	18.653	38.626	0.647
G-SOMP+	7.380	40.418	0.790	9.767	27.530	0.487
CNN-Fus	16.911	39.248	0.594	17.525	29.592	0.260
CNBM	**2.338**	**53.149**	**0.983**	4.784	**41.174**	**0.885**

TABLE III
FUSION EVALUATION RESULTS OF RESISTING REGISTRATION ERROR (SSIM)

	Number of pixels dislocation				
	4	8	12	16	20
CNMF	0.729	0.479	0.395	0.324	0.304
NLSTF_SMBF	0.756	0.757	0.758	0.757	0.759
GSA	0.663	0.586	0.464	0.386	0.412
Hysure	0.687	0.505	0.349	0.230	0.149
FUSE	0.642	0.530	0.439	0.355	0.313
OTD	0.721	0.726	0.719	0.721	0.715
G-SOMP+	0.672	0.683	0.675	0.671	0.674
CNN-Fus	0.451	0.357	0.332	0.319	0.314
CNBM	**0.941**	**0.945**	**0.942**	**0.948**	**0.945**

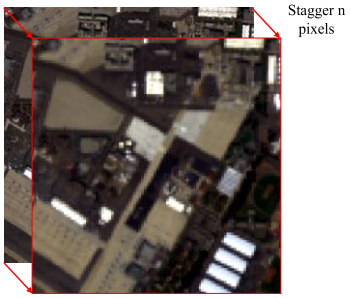

Fig. 4. Schematic of image registration shift.

deciding fusion effect and cannot be ignored in all situations; thus, the followed experiments are necessary. To simulate the geographic registration dislocation between HRMS and LRHS images, different pixels offsets are set, including 4, 8, 12, 16, and 20 pixels offsets. The construction of the dislocation process is shown in Fig. 4, where the dislocation is carried out according to the diagonal of the image.

For each group of simulated dislocation datasets, nine algorithms are used to fuse. The fusion evaluation results are shown in Table III. In order to evaluate the fusion results from both two dimensions of spatial and spectral, the SSIM is selected as the evaluation index to evaluate the fusion performance in this group of experiments.

From Table III, it can be easily observed that the CNBM method can effectively keep the fusion accuracy along with registration error increasing. The NLSTF_SMBF, OTD, and G-SOMP+ can also resist a registration error to some extent because some DicL techniques are used in these three algorithms. However, since none of them adopt any nonoverlapped bands construction measures, the holistic fusion results in the first dataset are not ideal. Besides, the evaluation index of other five algorithms shows a significant downward trend with registration error increasing. This is also a typical shortcoming of traditional pixel-to-pixel correspondence HMF methods.

The fusion results with true color composition pattern are shown in Fig. 5 to intuitively display the texture information of fusion results under different degrees of image registration error. As shown in Fig. 5, we can observe that the fusion results of CNMF, GSA, Hysure, FUSE, and CNN-Fus algorithms are gradually blurred with registration error increasing. Even the fusion result of FUSE and CNN-Fus has appeared serious ghosting phenomenon. Among them, CNN-Fus and Hysure are the most serious. The main reason for this situation is that the pixel-to-pixel registration-based HMF methods have high requirements for geographic registration. Once there appears

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Fig. 5. Fusion results of nine algorithms shown in true color.

dislocation, the HRMS image cannot match the exact position with the LRHS image, which will largely affect the fusion accuracy. As it can be seen in the last column, the proposed method maintains better image texture and color information. The NLSTF_SMBF, OTD, and G-SOMP+ also perform well in keeping spatial information fidelity. However, combined with Table III, these three algorithms do not achieve better fusion results in spectral information reconstruction. To make spectral fusion performance better visualization, SAM error images are given in Fig. 6. As we can see in Fig. 6, the color bar in the reference image shows that the dark blue indicates that the spectral information is well preserved. Conversely, the bright yellow represents that the spectral information is not reconstructed accurately. It can be analyzed by combining Fig. 6 that the proposed method performs well in spectral information fusion. The spectral fusion accuracy of CNMF, GSA, Hysure, and FUSE algorithms presents a downward trend with registration error increasing. This conclusion is consistent with that obtained from Fig. 5.

B. Validation Experiments on ZY1E Images

Ziyuan-1 02D (ZYIE) satellite, also known as 5-m optical satellite, carries both MS and HS sensors simultaneously. The spatial resolution of MS sensor of ZY1E is 10 m with a 115 km width. It has eight spectral bands and a range from 452 to 1047 nm. The spatial resolution of HS sensor is 30 m with 60 km width. Its spectral bands range from 400 to 2500 nm with totally 145 bands. We select two images captured on July 29, 2021 and a part of the overlapped region is cropped as the second experimental dataset in this article. The size of this cropped area is 2700 $\times$ 2700 pixels. The corresponding HRMS and LRHS images of ZY1E dataset are shown in Fig. 7. The fundamental HS and MS curves of the major land-cover objects in the ZY1E dataset are shown in the last column of Fig. 7.

Due to the lack of real HRHS image in this dataset, the quantitative fusion performance evaluation is absent. Real RGB composite image, detailed subimage, and standard false color composite of nine fused images and reference image are presented to show the fusion results.

In Fig. 8, the reference image is RGB composite of HRMS image, and the land-cover object texture is obvious. The images in Fig. 8(a)–(i) are the fusion results of the comparison methods. From the results, the fused image of NLSTF_SMBF and G-SOMP+ algorithm has a large color distortion compared with a reference image, which indicates that these two methods produce a large spectral distortion; in terms of overall fusion images definition, Hysure, OTD, and CNN-Fus algorithms cannot maintain texture feature well. The fusion results of them are blurred, especially in the roads of this scene; in Fig. 8(i), it can be found that texture and color of the proposed method achieve most similar compared with reference image.

In addition, in Fig. 9, two subimages are selected to display detailed texture information of the fused images by different comparison methods. For satellite data, the spatial resolution

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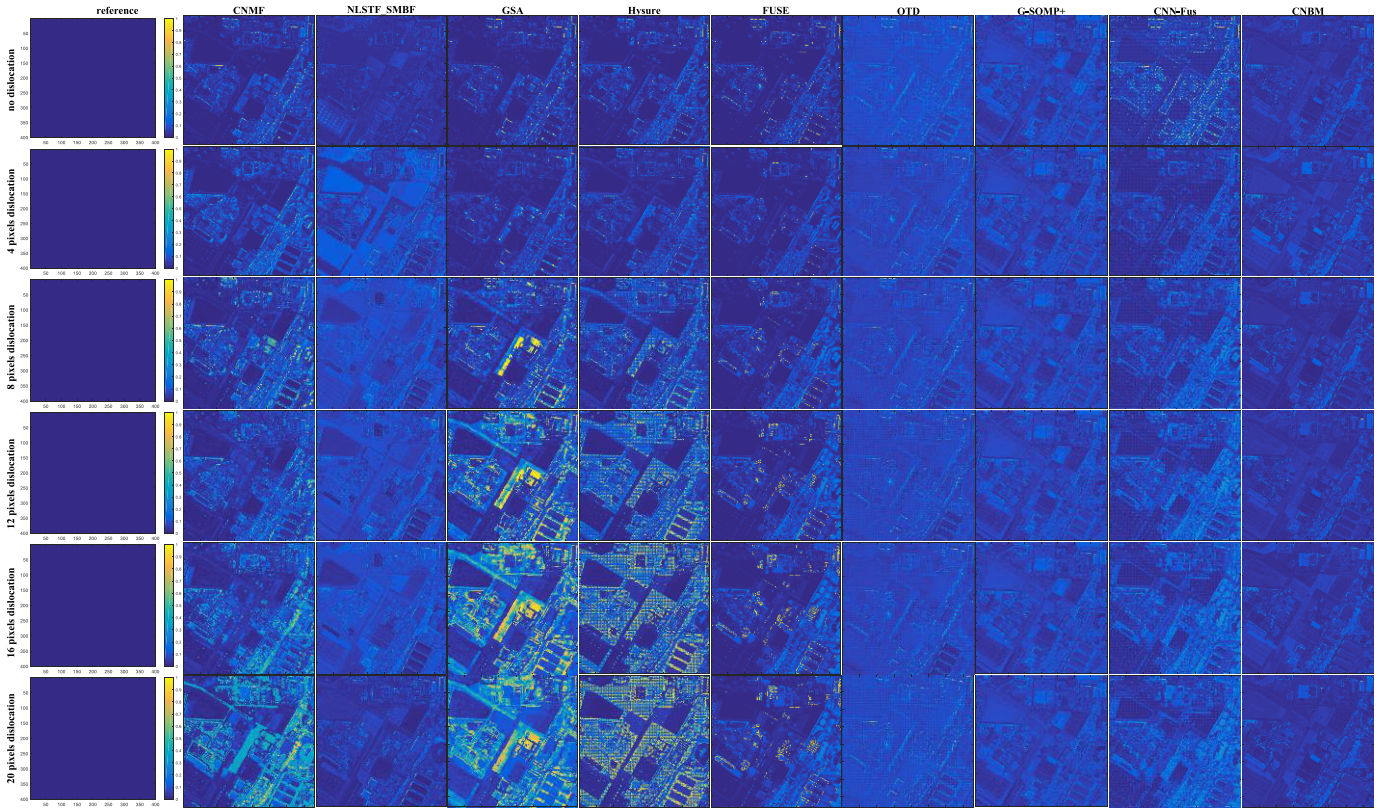

Fig. 6. SAM error images of fusion results using nine algorithms.

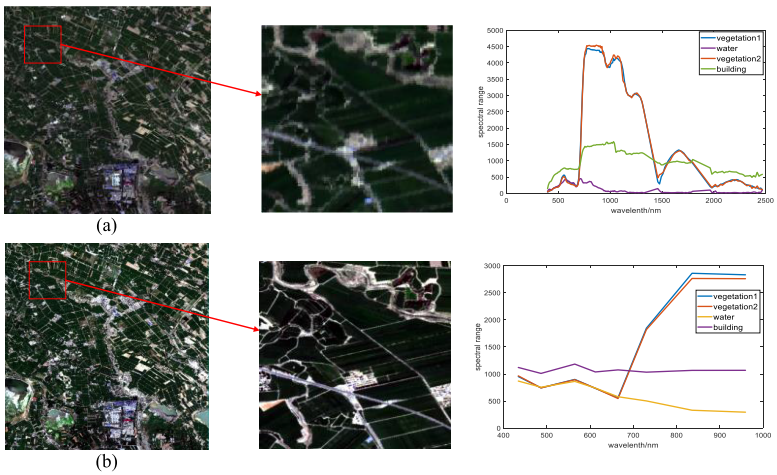

Fig. 7. (a) LRHS and (b) HRMS image in the ZYIE dataset.

is determined by image acquisition pattern and sensor. The spatial resolution of subsatellite points and non-subsatellite points is different [39]. Resampling original image could bring systematic registration error. The larger the area, the more serious the situation. This is also the case with ZY1E remote sensing image. Thus, it can be observed from Fig. 9 that the fusion results of HMF methods only adapting pixel-to-pixel fusion coregistering strategy have different degrees of pixel dislocation, and even there does not add any artificial error. This is related to the imaging pattern. As it can be seen in Fig. 9, the lines of roads and edges of fields in fused subimages by CNBM are very clear. It is obvious that the proposed method can better solve the registration error problem in the HMF situation. Moreover, the hue of image is also the most similar to the reference image, which shows that the proposed method can maintain spectral information when there exist systematic registration errors.

The images in Fig. 10 present the fusion results with standard false color composite. The reference image is LRHS image shown in the standard false color composite. Since vegetation occupies a large percent of area in this dataset, therefore, whether the fused image can improve the vegetation information is important. In Fig. 10, it is more intuitive to show the accuracy of red edge spectral

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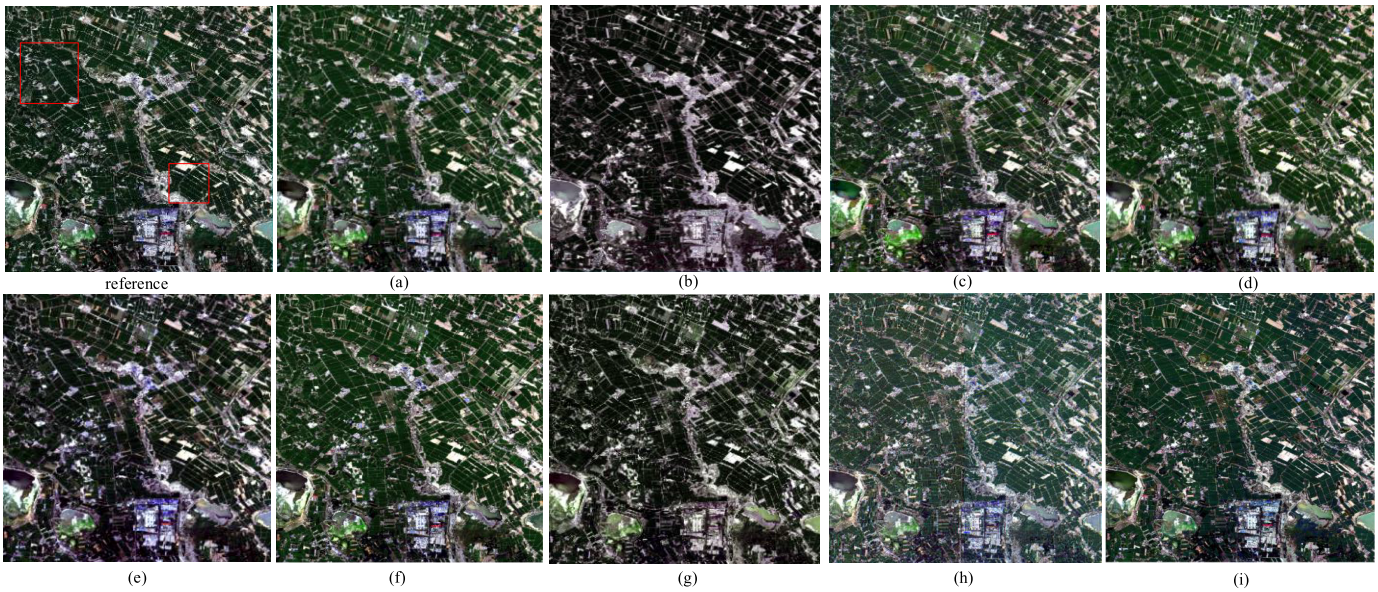

Fig. 8. Fusion result of nine algorithms in the ZY1E dataset. (a) CNMF. (b) NLSTF_SMBF. (c) GSA. (d) Hysure. (e) FUSE. (f) OTD. (g) G-SOMP+. (h) CNN-Fus. (i) CNBM.

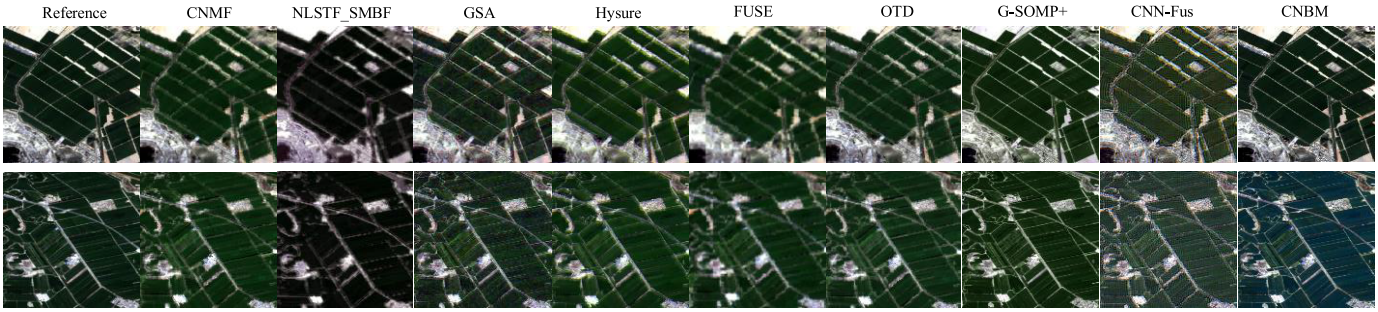

Fig. 9. Comparison of some image details of nine fusion methods in two subareas of ZYIE dataset. (First column) Subarea of HRMS. (Remaining columns) Two subareas in Fig. 8(a)–(i), respectively.

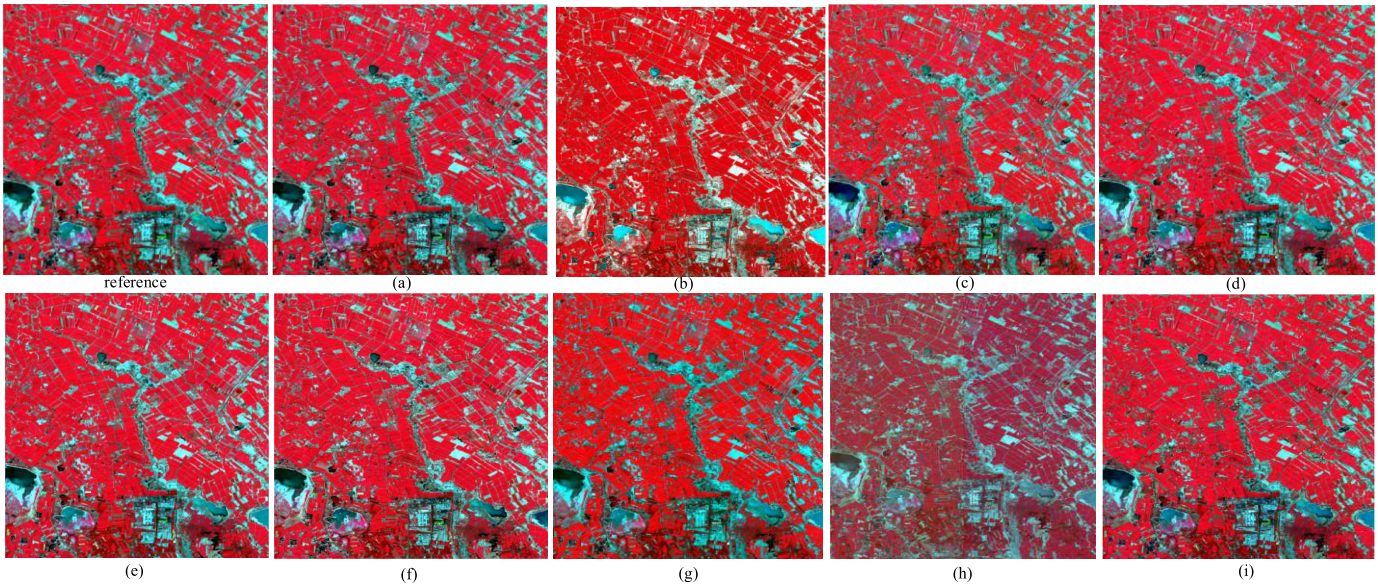

Fig. 10. Fusion result of nine algorithms in the ZY1E dataset using standard false color composite. (a) CNMF. (b) NLSTF_SMBF. (c) GSA. (d) Hysure. (e) FUSE. (f) OTD. (g) G-SOMP+. (h) CNN-Fus. (i) CNBM.

bands reconstruction. It can be found that the fusion results of NLSTF_SMBF, G-SOMP+, and CNN-Fus methods produce a color shift compared with the reference image; the fusion results of CNMF, GSA, Hysure, FUSE, OTD, and CNBM methods have similar color details to reference image, which indicates that these six algorithms can

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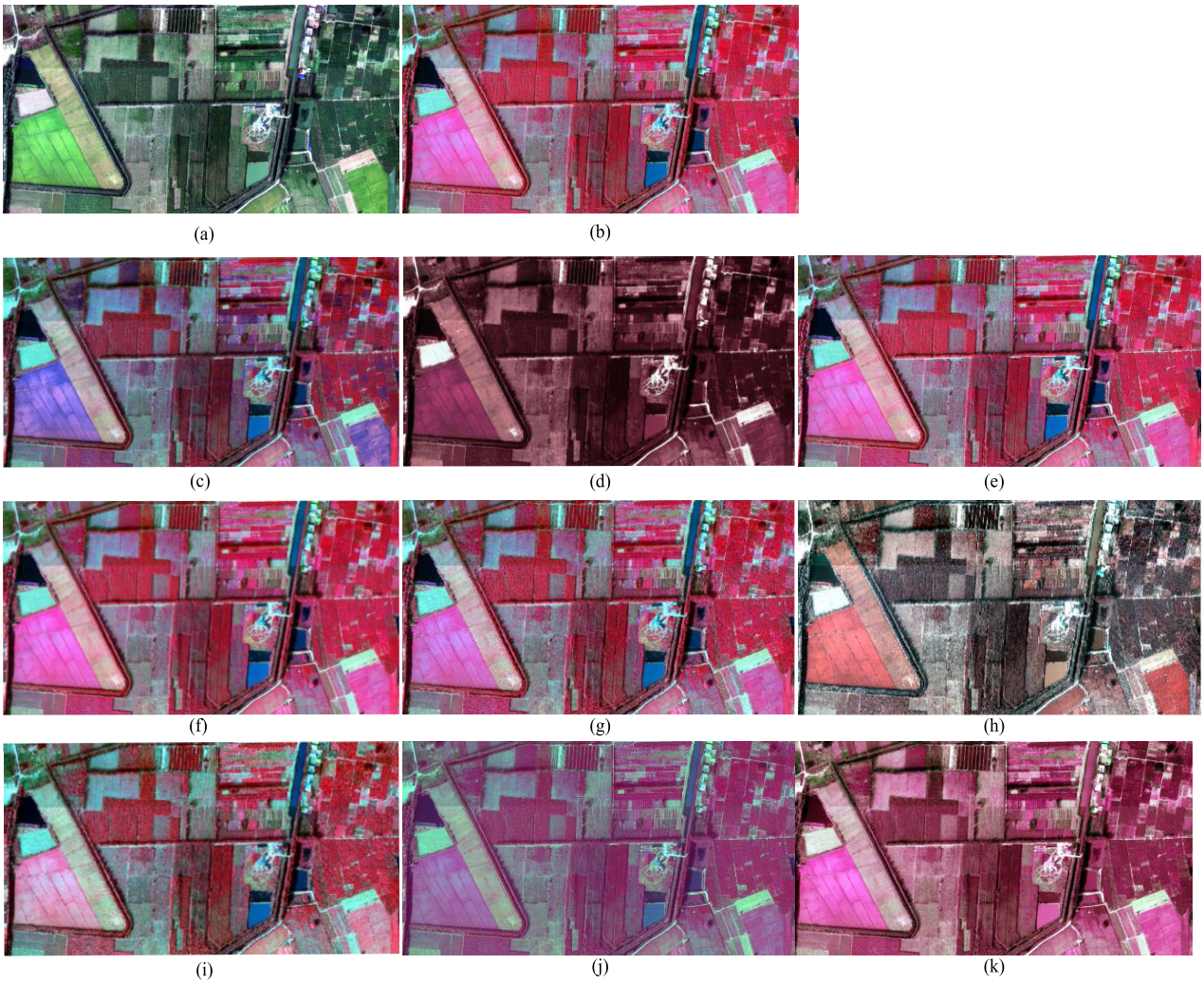

Fig. 11. (a) RGB composite and (b) standard false color composite of XiongAn dataset. (c)–(k) Fusion results of nine fusion algorithms for standard false color composite. (c) CNMF. (d) NLSTF_SMBF. (e) GSA. (f) Hysure. (g) FUSE. (h) OTD. (i) G-SOMP+. (j) CNN-Fus. (k) CNBM.

recover red edge spectral information correctly in the ZY1E dataset.

C. RGB and HS Image Fusion on Simulated UAV Image

The RGB image can be seen as a special kind of MS image [5]. Thus, the proposed method can also solve RGB and HS image fusion problems. As shown in Fig. 11, the UAV image is obtained in Xiong'an Hebei province in July 2021 with a cropped spatial size of 1500 × 3000 pixels, and its spatial resolution is 1.3 m. The spectral range of this UAV HS image is in the range of 400–1000 nm with a total of 256 bands. Its spectral resolution is within 2 nm. A pair of RGB and LRHS images are simulated from the original UAV HS image. The spatial resolution of LRHS is three times downsampled from the original HRHS image, that is, 3.6 m. Thus, the size of simulated LRHS is 500 × 1000 pixels. Jiang et al. [47] provided SRF of 28 different types of cameras, and the used SRF to simulate RGB is selected from it at random.

Fig. 11(a) and (b) shows the RGB and standard false color composite of XiongAn dataset, respectively. Fig. 11(c)–(k) shows the standard false color composite of the fusion results. From intuitive visual point of view, most of the fusion results in the XiongAn dataset are not ideal. The reasons for this phenomenon are that available matching spectral bands of RGB image are fewer than the MS image. The probability of spectral distortion increases. From the perspective of intuitive color fidelity, during nine HMF methods, GSA, Hysure, and FUSE algorithms produce better results than other methods. However, from quantitative evaluation results (refer to Table IV), GSA, Hysure, and FUSE do not maintain good texture information because PSNR and SSIM indexes are not very well. In particular, there is a registration error, and all evaluation indexes drop evidently. The fusion results of CNMF, NLSTF_SMBF, OTD, and CNN-Fus methods have large-area color shift. Partial area in fused image by CNMF appears blueshift. The fusion result of NLSTF_SMBF produces a relative largely spectral distortion. The OTD algorithm does not persist the red edge information well. For the CNBM method, the area of largest spectral distortion is the water area. Affected by phytoplankton, the visible spectral curve of water in this dataset is similar to that of vegetation, which produces

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TABLE IV
EVALUATION RESULTS OF NINE ALGORITHMS ON THE XIONGAN DATASET

	no registration dislocation			3 pixels dislocation			Time/s
	SAM(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)	SAM(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)	
CNMF	3.356	35.189	0.752	4.786	32.589	0.696	2262
NLSTF_SMBF	4.832	30.509	0.665	4.790	30.579	0.666	661
GSA	**2.101**	**38.248**	0.770	**3.079**	34.683	0.627	80
Hysure	2.469	35.258	0.644	3.287	32.941	0.591	6052
FUSE	2.761	32.426	0.629	3.674	29.137	0.312	47
OTD	10.910	35.363	0.585	11.194	31.274	0.426	6067
G-SOMP+	9.237	33.293	0.592	9.283	30.284	0.483	4820
CNN-Fus	13.404	28.417	0.410	14.913	25.115	0.182	1801
CNBM	5.843	34.069	**0.782**	5.763	**35.548**	**0.778**	1228

TABLE V
FUSION EVALUATION RESULTS OF OVERLAPPING AND NONOVERLAPPING SPECTRAL BANDS

	Overlapped spectral range			Non-overlapped spectral range		
	SAM(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)	SAM(↓,0)	PSNR(↑,+∞)	SSIM(↑,1)
CNMF	5.262	42.469	0.858	3.848	22.233	0.499
NLSTF_SMBF	6.253	30.285	0.742	5.973	20.374	0.419
GSA	4.495	43.167	0.852	**2.213**	26.665	0.461
Hysure	**3.676**	40.539	0.766	2.468	25.337	0.412
FUSE	4.328	35.559	0.410	2.714	22.866	0.226
OTD	15.174	37.614	0.732	12.363	25.545	0.534
G-SOMP+	11.851	27.052	0.602	8.933	18.075	0.443
CNN-Fus	19.898	23.017	0.412	16.314	10.040	0.204
CNBM	4.321	**49.088**	**0.914**	3.750	**29.895**	**0.654**

nonnegligible influence of fusion result. Besides, other region of CNBM fusion result is relatively coincident with reference image.

Table IV shows the evaluation results of nine HMF algorithms in the XiongAn dataset. Meanwhile, to evaluate the registration error resistance ability and calculation cost, a group of comparative experiments are added, in which 3 pixels registration error is simulated between RGB and LRHS image, that is, a registration error of 3.6 m.

As shown in Table IV, the GSA algorithm performs best when there is no registration error. The SAM and PSNR achieve optimal among nine HMF methods. This situation is different from that in the first dataset. This is because the spectral range of HRMS and LRHS in the XiongAn dataset is relatively close. This conclusion further confirms that the smaller the spectral range difference of the images to be fused, the better the fusion performance of GSA algorithm, which is also a typical characteristic of CS-based HMF algorithms. However, when there exists a small registration error, three evaluation indicators of GSA decrease evidently. Accordingly, the fusion performance of Hysure, FUSE, and CNN-Fus methods also decreases when it has a small registration error. Similar to the conclusion obtained in the first dataset, the methods that utilize the DicL technique can better resist registration error, such as NLSTF_SMBF, OTD, and G-SOMP+. From the above analysis, we can conclude that the fusion performance of RGB and HS image is greatly affected by a registration error, but SR-based methods can relieve such situation. By comparison, a CNBM algorithm always keeps a stable and well fusion performance. In particular, there is a registration error, and the fusion performance of CNBM is little affected.

Table V shows the fusion evaluation results of overlapped and nonoverlapped spectral bands in the XiongAn dataset. In this dataset, the nonoverlapped spectral bands are located on the red edge area. The characteristics of this curve are relatively smooth and there is less feature peak and valley. Thus, the SAM index of nonoverlapped bands is higher than overlapped spectral bands. The PSNR is contrary. In the overlapped spectral range, the SAM of Hysure method acquires the best performance among nine HMF methods. However, the SAM of GSA algorithm in nonoverlapped bands achieves the best. For the other two fusion evaluation indexes, the PSNR and SSIM of the proposed method achieve the best in both overlapped and nonoverlapped bands.

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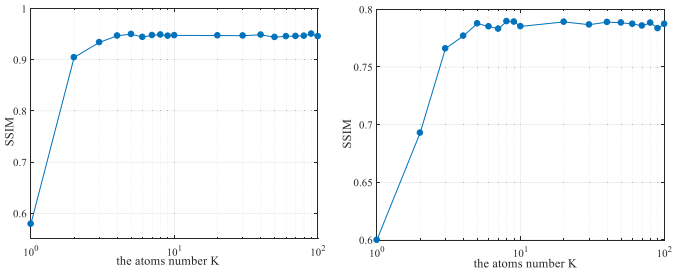

(a) (b)

Fig. 12. SSIM of CNBM under different K values in (a) SanDiego Airport and (b) XiongAn datasets.

D. Discussion on Parameter Setting

The fusion performance of the proposed method has highly correlated with parameter setting. The most important parameter in the proposed method is the number of atoms participated in reconstruction K. To explore the quantitative relationship between parameter setting and fusion performance, a series of experiments are designed under different parameter values.

The fusion evaluation index SSIM of the proposed method under different K values in two datasets is shown in Fig. 12(a) and (b). As can be observed in Fig. 12(a), the SSIM shows an upward trend with K increasing when K is less than 5. When K is greater than 5, the SSIM value presents a stable state with K changing. The same phenomenon can be observed in Fig. 12(b). In Fig. 12(b), it can be found that the SSIM achieves the best when K is set to 4. Before K reaches 4, SSIM and K maintain a positive correlation, and then, with K increasing, SSIM maintains a stable state. From the abovementioned relationship between SSIM and K value, we can find another interesting conclusion that the K value is almost the same as the number of ground objects in each dataset when SSIM achieves optimal. Therefore, we can take the number of ground objects in the to-be-fused images as the empirical value of K. As result, K is set to 5 in the SanDiego Airport, 4 in the XiongAn datasets, and 3 in the ZY1E dataset.

V. Conclusion

Data fusion can be regarded as an effective way to solve the spatial resolution limitation of HS image. In the process of image fusion, inevitable image registration error and nonoverlapped spectral information completion will affect the fusion performance greatly. To alleviate spatial registration error and reconstruct nonoverlapped spectral information, a novel HMF DicL framework is proposed, which considers nonoverlapped spectral bands reconstruction and image misregistration (CNBM). The proposed framework considers from the perspective of pure end-members matching instead of pixel-to-pixel registration. In this way, the unavoidable registration error in the pixel-based registration fusion method can be eliminated to great extent. Especially for satellite data, the spatial resolution of subsatellite point area and other area has a slight difference, which adds systematic error to geographical registration. Due to the different sensor of HS and MS images to be fused, the spectral range of two images may be partially different. Unlike most fusion methods, the proposed method separates a nonoverlapped spectral dictionary from an HS dictionary to represent the subspace of nonoverlapped spectral bands. Using nonoverlapped and entire HS dictionaries, the quantitative relationship of nonoverlapped bands and entire fusion bands can be calculated. Finally, the entire spectral range can be fused accurately. To evaluate the fusion performance of the proposed method, three different cases of datasets were used to test, including simulated and real HMF fusion test dataset, RGB, and LRHS fusion test dataset. All of them have incomplete overlapped spectral range. The traditional fusion and registration error resistance performance are validated in the experimental section. The experimental results show that the proposed method can effectively solve problems of image misregistration and nonoverlapped spectral bands reconstruction.

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